



Quantum computing

Taras Holotyak

Content of course

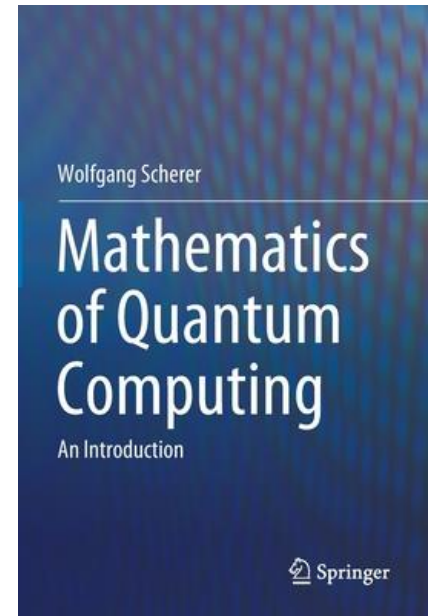
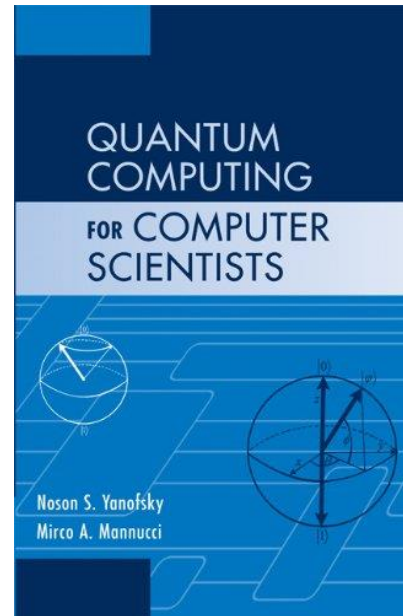
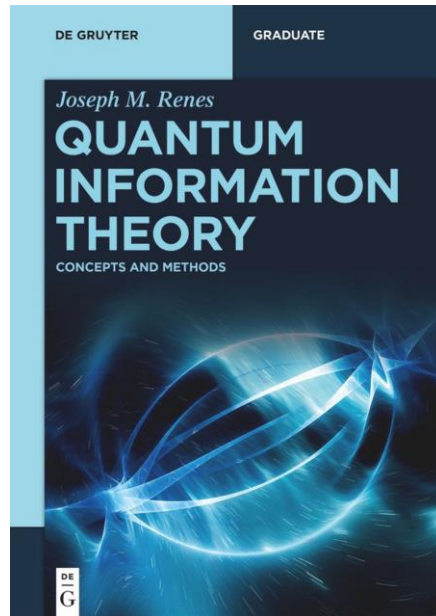
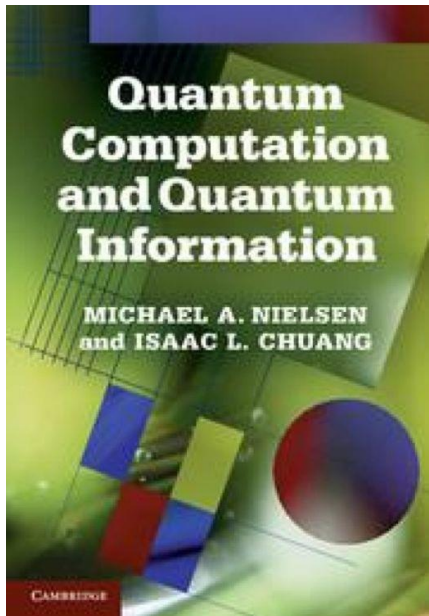
- Theoretical basis for quantum computing.
- Practical basis for quantum computing.
- Near-term potential of quantum computing.
- Other applications of quantum information.

Course grades

Lectures (Oral exam): 50%

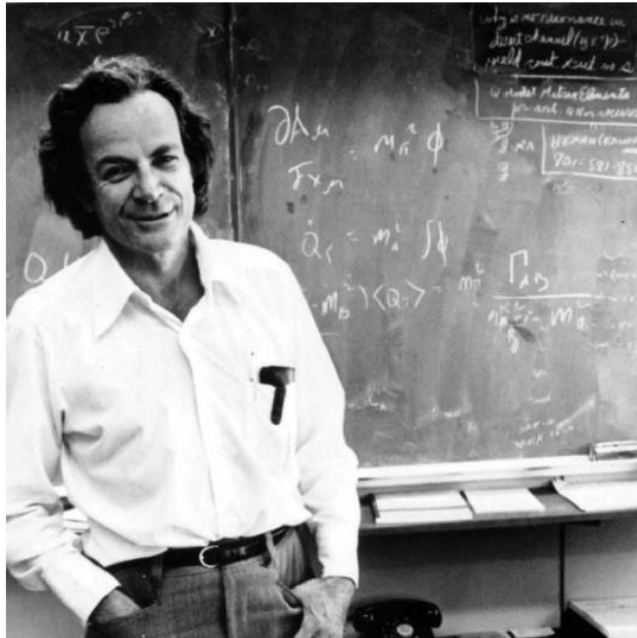
TP: 50%

Recommended books



Introduction to quantum computing

What is quantum computing?



<https://it.wikipedia.org/wiki/File:Richard-feynman.jpg>

. . . trying to find a computer simulation of physics seems to me to be an excellent program to follow out. . . . the real use of it would be with quantum mechanics. . . . Nature isn't classical . . . and if you want to make a simulation of Nature, you'd better make it quantum mechanical, and by golly it's a wonderful problem, because it doesn't look so easy.

Richard Feynman

Introduction to quantum computing

What is quantum computing?



<https://quillette.com/2019/01/26/the-unconstrained-vision-of-david-deutsch>

The theory of computation has traditionally been studied almost entirely in the abstract, as a topic in pure mathematics. This is to miss the point of it. Computers are physical objects, and computations are physical processes. What computers can or cannot compute is determined by the laws of physics alone, and not by pure mathematics.

David Deutsch

Introduction to quantum computing

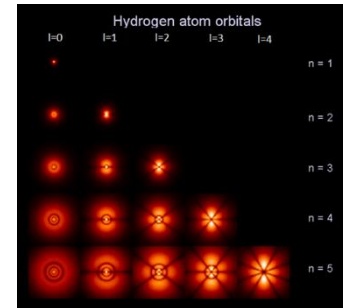
What is quantum computing?

It is computation (and more generally, information processing) based on the principles of **quantum mechanics**, rather than classical physics.

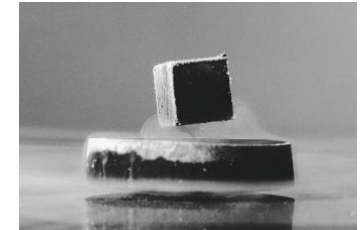
Quantum mechanics is a description of nature at its most fundamental level.

- Formulated in the early 20th century to explain the behavior of subatomic particles.
- QM, and its specializations (quantum field theory, quantum chromodynamics, etc), have been spectacularly successful in explaining microscopic physical phenomena.

Hydrogen atom



Superconductivity



Introduction to quantum computing

Why study quantum computing?

Article

Quantum supremacy using a programmable superconducting processor

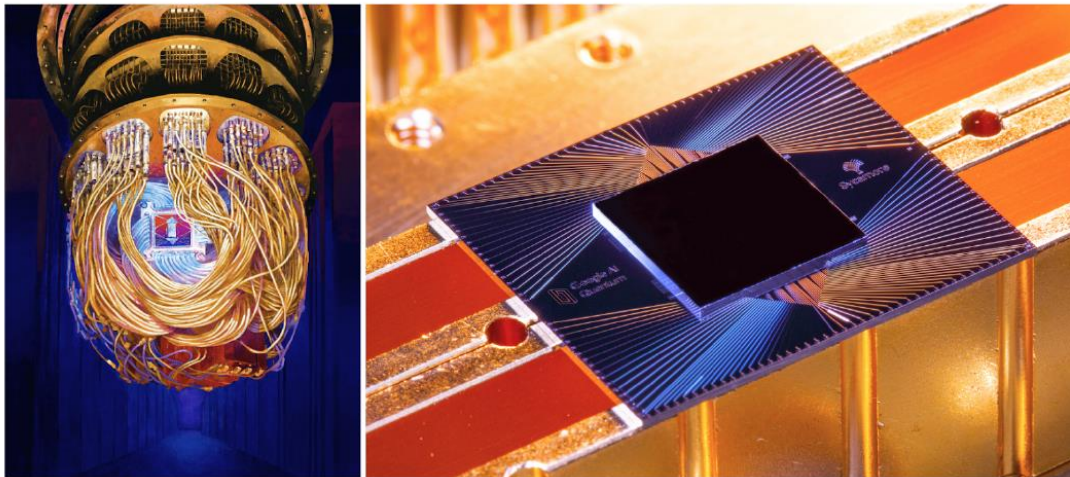
<https://doi.org/10.1038/s41586-019-1666-5>

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Frank Arute¹, Kunal Arya¹, Ryan Babbush¹, Dave Bacon¹, Joseph C. Bardin^{1,2}, Rami Barends¹, Rupak Biswas³, Sergio Boixo¹, Fernando G. S. L. Brandao^{1,4}, David A. Buell¹, Brian Burkett¹, Yu Chen¹, Zijun Chen¹, Ben Chiaro⁵, Roberto Collins¹, William Courtney¹, Andrew Dunsworth¹, Edward Farhi¹, Brooks Foxen^{1,5}, Austin Fowler¹, Craig Gidney¹, Marissa Giustina¹, Rob Graff¹, Keith Guerin¹, Steve Habegger¹, Matthew P. Harrigan¹, Michael J. Hartmann^{1,6}, Alan Ho¹, Markus Hoffmann¹, Trent Huang¹, Travis S. Humble⁷, Sergei V. Isakov¹, Evan Jeffrey¹,



<https://ai.googleblog.com/2019/10/quantum-supremacy-using-programmable.html>

Introduction to quantum computing

Why study quantum computing in computer science?

Quantum software development

Quantum algorithm design

Quantum machine learning

Quantum computer architecture and compiler design

Quantum information theory research

Quantum communication system development

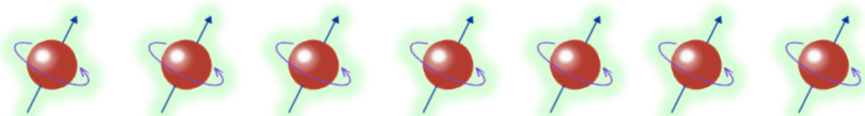
Quantum security system development

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The exponentiality of QM

After nearly a century of study, the best (classical) methods for predicting the behavior of general quantum systems require exponential resources.

- The state of N particles requires at least 2^N numbers to describe.


$$|\psi\rangle = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_{2^N} \end{pmatrix} \in \mathbb{C}^{2^N}$$

- For $N \sim 300$ (the number of particles in a single uranium atom),
 $2^N \gg \#$ of atoms in observable universe $\approx 10^{82}$

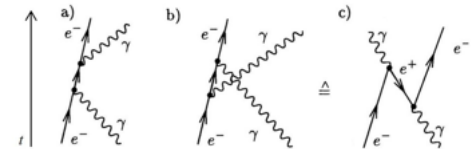
Nature seems to be doing extravagant amounts of computation “behind the scenes”!

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The exponentiality of QM

- So how do physicists actually do (quantum) physics?
 - **Answer #1:** 100+ years of extremely clever approximations and shortcuts for calculations.

- Bethe ansatz, Feynman diagrams, perturbation theory, mean field approximations,...



- **Answer #2:** computer simulations
 - Density functional theory, Quantum Markov Chain Monte Carlo,...
- **Answer #3:** studying systems that allow for Answers #1 and #2 above
 - Systems that are mostly classical
 - Systems with non-strongly interacting particles
 - Lucky that most things we've studied so far fall in this category!
- But simulating *general* quantum physics still seems like a hard task...



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The exponentiality of QM

- In 1981 Richard Feynman asked the following question:
“Can probabilistic computers simulate quantum mechanics?”
- His conclusion: NO!
- *“...Nature isn’t classical, damnit, and if you want to make a simulation of nature, you’d better make it quantum mechanical.*
- *By golly, it’s a wonderful problem, because it doesn’t look so easy.”*
- Feynman (along with David Deutsch, Paul Benioff, Yuri Manin) came up with the idea of computing that is fundamentally based on quantum mechanics.



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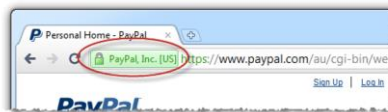
Early days of QC

- Feynman's motivation for quantum computers is the most obvious one: **simulating quantum systems**. (almost tautological!)
- This was a radical idea, and not appreciated for many years.
- 1980s: David Deutsch found example of a (non-physics) problem that could be solved faster on QC than any classical computer (by a constant factor)
- 1992: Bernstein and Vazirani, and then Dan Simons found examples of (non-physics) problems that could be solved by QC **super-polynomially faster** than with classical computer.
- 1994: Peter Shor showed up...

$$P = 17449$$

$$Q = 34253$$

$$P \times Q = 597680597$$



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Early days of QC

- 1994: Peter Shor realized that Simons's idea could be extended to efficiently solve the following problem:
- **(Integer Factorization)** Given integer $N > 0$, output its prime factorization.
- The security of the RSA cryptosystem (the most widely deployed public-key encryption on the internet) crucially relies on the **assumption that factoring integers is hard!**



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Early days of QC

Crossroads

Since Shor's algorithm, physicists and computer scientists have been faced with three options:

1. Quantum mechanics is wrong.
2. There is a fast classical algorithm for factoring.
3. There must be a fundamentally more powerful model of computation than classical computers.

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QC rumour

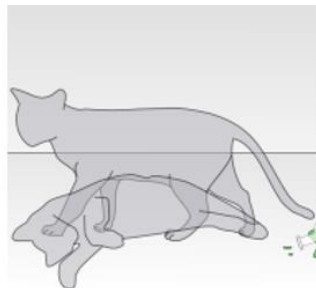
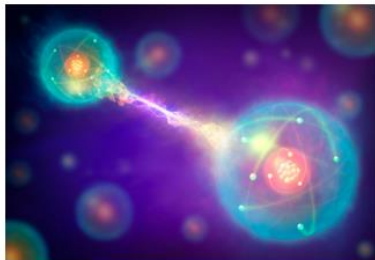
Myth: Quantum computers solve things by trying every possibility at once.

FALSE

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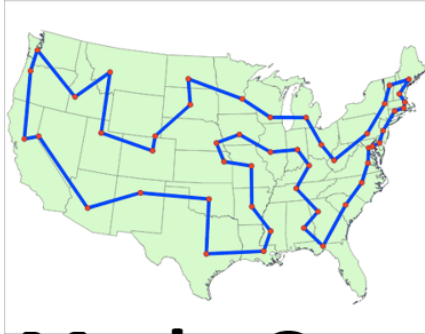
QC rumour

Fact: Quantum computers can solve certain types of problems faster by carefully leveraging quantum effects such as entanglement, interference, superposition....



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QC rumour



Myth: Quantum computers can solve the traveling salesman problem quickly.

FALSE

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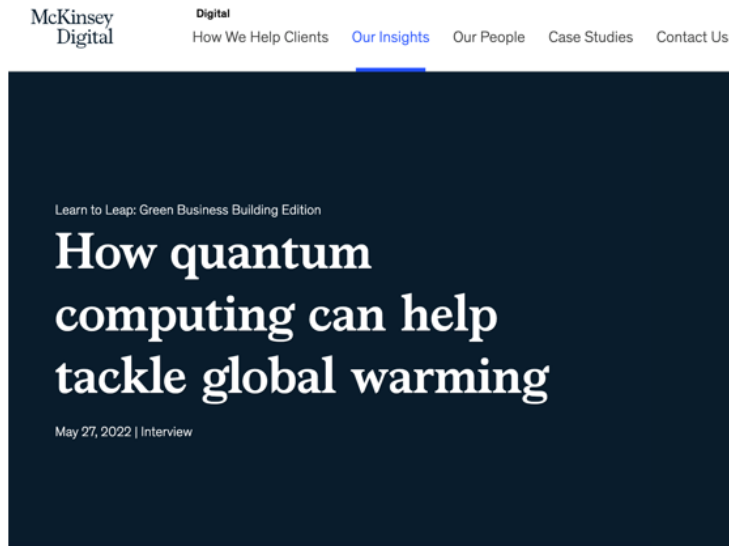
QC rumour

Fact: Quantum computers *probably* cannot solve TSP quickly.

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QC rumour

Claim: Quantum computers will solve climate change.



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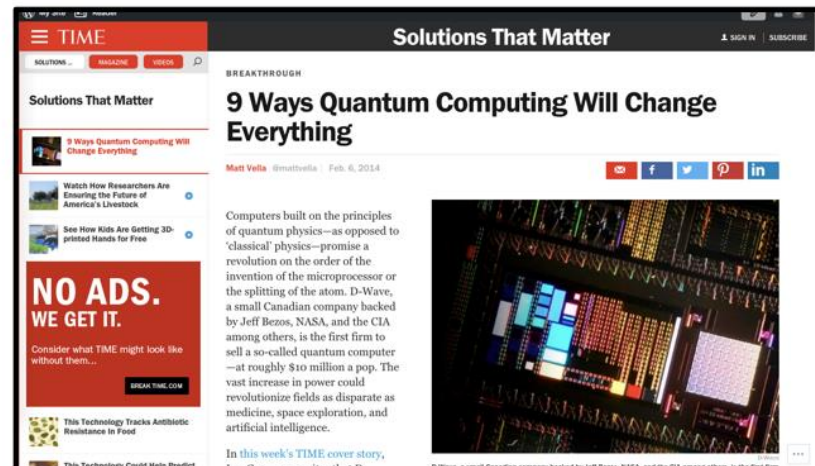
QC rumour

Unclear. Quantum computers *might* solve hard chemistry/physics problems that *might* be related to designing more sustainable materials, batteries, technologies.

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QC rumour

Claim: Large-scale quantum computers with millions of qubits are just right around the corner.



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QC rumour

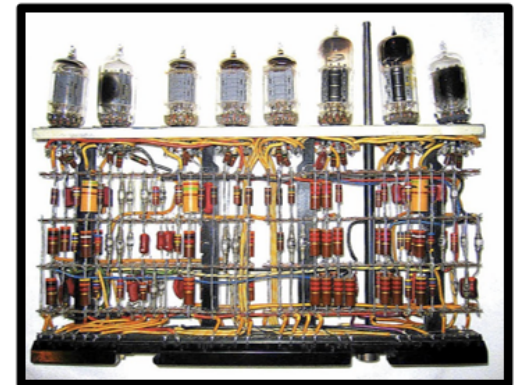
Exciting times for quantum computing:

1. **Engineering**: increasingly sophisticated quantum computers are being built, with several hundred controllable qubits.
2. **Theoretical developments**: new algorithms, new cryptographic protocols, new insights into how quantum computing compares to classical computing
3. **Deep connections to other sciences**: condensed matter physics, quantum gravity, materials science, chemistry, computer science, pure mathematics.

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Emerging quantum computers

- Until 2017 or so, most quantum computing devices had ~ 10 qubits.
- Suddenly, there was a flood of resources devoted to engineering efforts, and now we are seeing programmable quantum computers with ~ 200 qubits.
- Major problem: *devices are insanely noisy!*
- We have been in the “**NISQ**” era:
 - Noisy Intermediate-Scale Quantum
 - Noisy devices with ~ 100 qubits.
 - Not capable of running Shor’s algorithm, but should still be capable of solving interesting, hard problems.



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Road of scaling

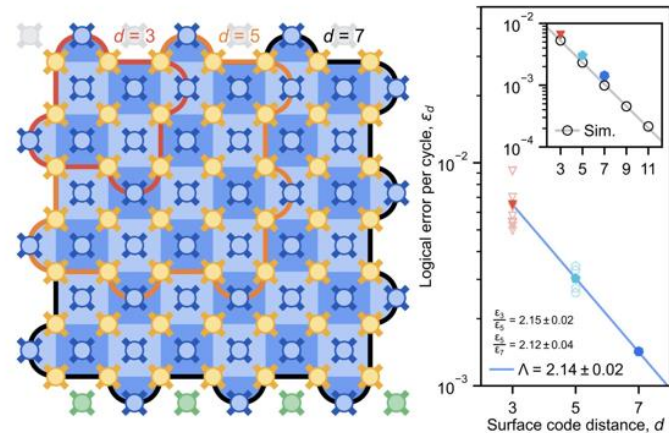
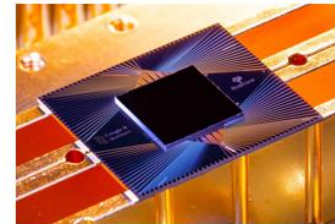
- In order to continue scaling up QCs, we need to make qubits more reliable.
- Pure hardware improvements alone won't suffice*; instead we need **quantum error correction**: a way to handle noise and errors in quantum systems.
- In order to get a **fault-tolerant QC**, we need to correct errors *faster than they occur*.
- In the last couple of years or so: incredible breakthroughs in error correction technology.

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Road of scaling

Quantum computing with superconducting qubits

- Qubits are superconducting loops of wire on a silicon chip.
- Platform of choice for Google, IBM, and many others.
- It was demonstrated an impressive error correction experiment with 101 physical qubits.
- The effective error rate *decreased* each time the code size increased.

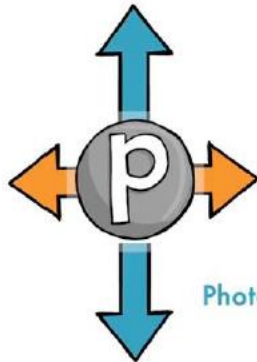


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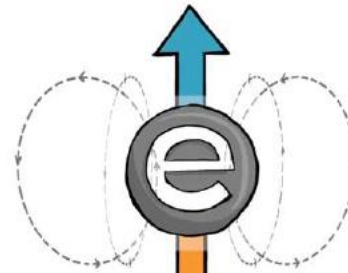


Persistent current in a superconducting circuit

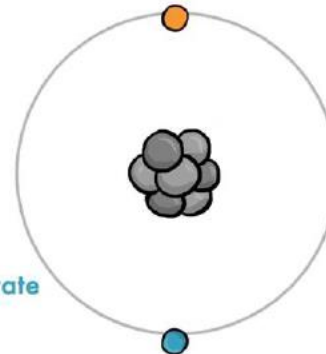


Photon polarization

QUBIT



Electron Magnetic Field



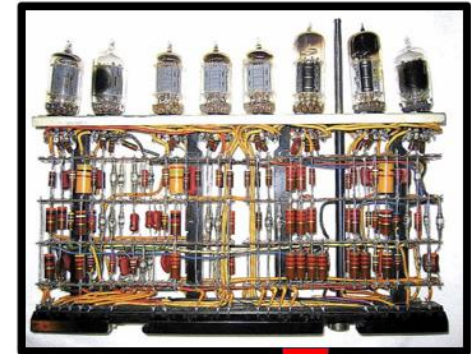
Atom Internal State

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Summary of hardware efforts

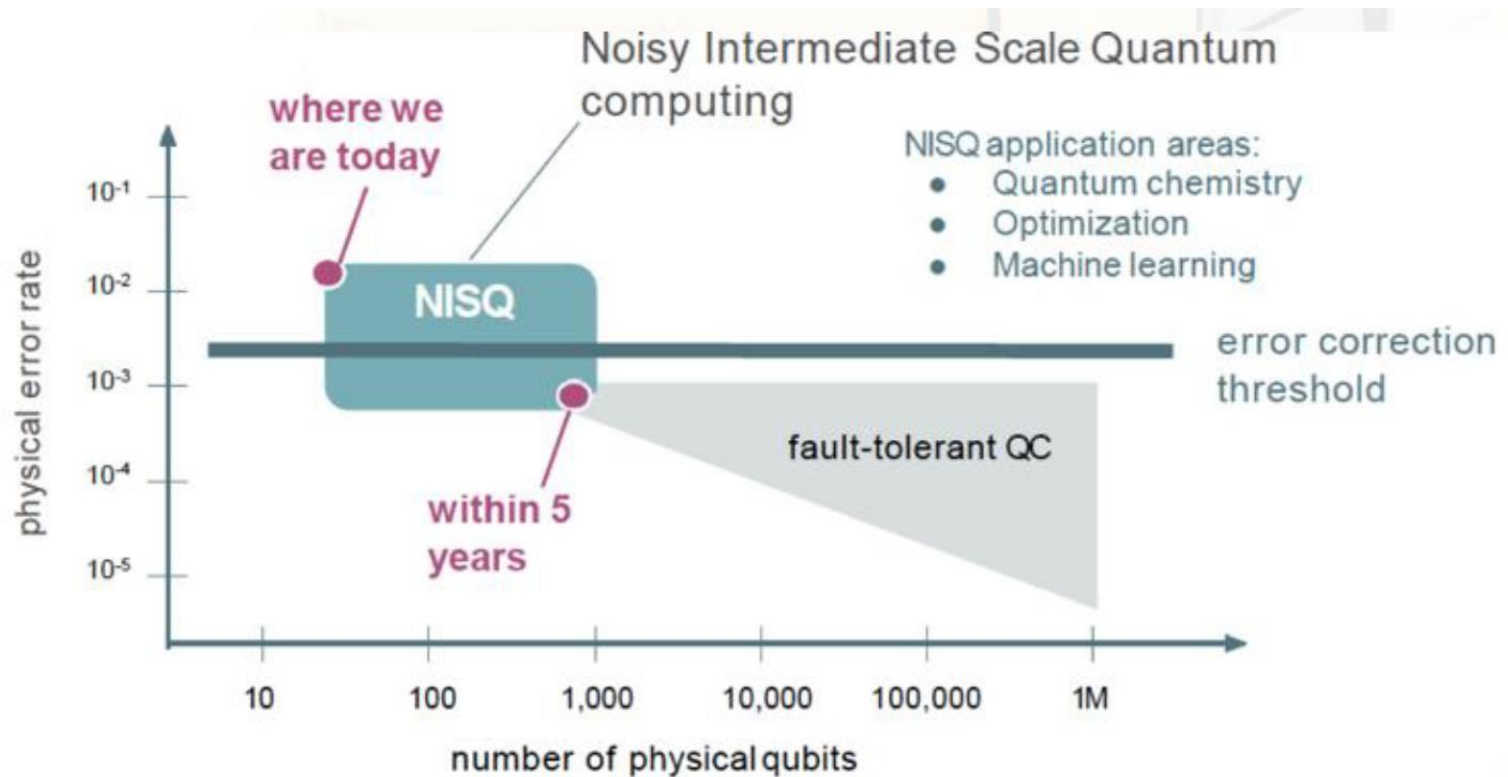
- We have been in the NISQ era, but it seems like we're at an inflection point.
- There's a "**qubit race**" going on, but qubit count is not everything!
- **Scaling up** (i.e. more qubits, better qubits) is a tough engineering challenge, but no fundamental obstacles to building QC with 10^6 noiseless qubits.
- Signs that quantum error correction is feasible and lots of exciting progress to come soon.



How can QCs be scaled up in a fault tolerant manner?

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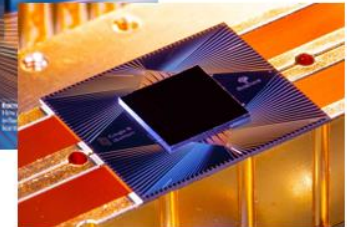


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Quantum supremacy

- In Fall 2019, Google announced that they had achieved “**Quantum Supremacy**” using their 53-qubit Sycamore quantum processor.
- **Quantum supremacy**: a convincing real-world demonstration of that a quantum computer is accomplishing a task that cannot feasibly be performed by a classical computer.
- Sweet spot: 53 qubits just beyond the reach of Google’s compute clusters to simulate in a reasonable time. ($\sim 2^{53}$ ops)

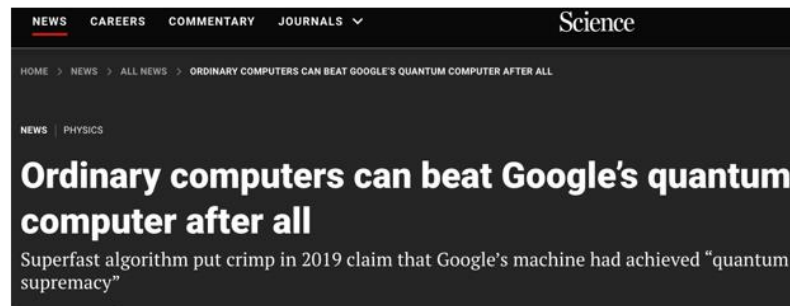
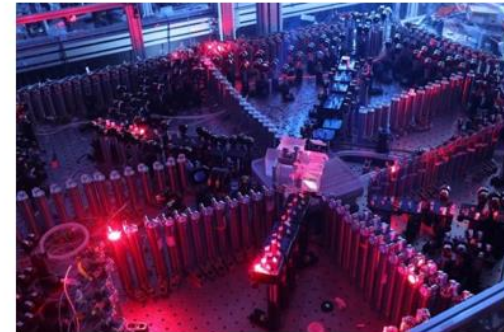


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Quantum vs classical

- Other research groups have also announced quantum supremacy results.
- Biggest competitor to quantum computers are *clever classical algorithms*:
 - In Nov 2021, a group from Chinese Academy of Sciences used tensor network algorithms to reproduce the statistics of the Google supremacy experiment using 512 GPUs in 15 hours.

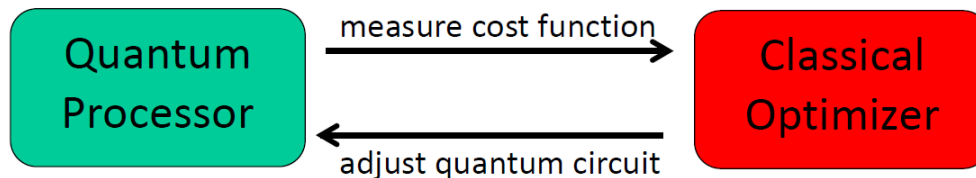


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Hybrid quantum/classical optimizers

Eddie Farhi: "Try it and see if it works!"



We don't expect a quantum computer to solve worst case instances of NP-hard problems, but it *might* find better approximate solutions, or find them faster.

Classical optimization algorithms (for both classical and quantum problems) are sophisticated and well-honed after decades of hard work.

We don't know whether NISQ devices can do better, but we can try it and see how well it works.

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Quantum-inspired classical algorithms

One of the most important problems in data-science is the construction of *recommendation systems*. Suppose that we have n products and a purchase history of m users, from which we need to make product recommendations:

- Until 2016 only techniques which run in time $\mathcal{O}(\text{poly}(mn))$ were known.
- In 2016 Jordan Kerenidis and Anupam Prakash published a quantum algorithm to achieve this task in time $\mathcal{O}(\text{poly log}(mn))$. That is, an exponential speed-up.
- Then in 2018 Ewin Tang published a classical algorithm inspired by Kerenidis and Prakash's quantum algorithm that also achieves the task in time $\mathcal{O}(\text{poly log}(mn))$.

Therefore, even in a classical world, there is merit in thinking quantumly.

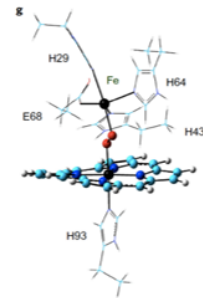
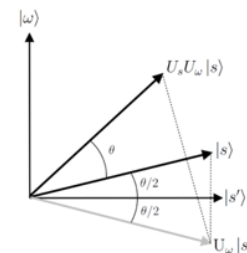
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Quantum algorithms

- Exponential speedups for structured, algebraic problems
 - Factoring (Shor's algorithm)
 - Hidden subgroup problem
- Polynomial speedups for unstructured search problems
 - Grover search
- Exponential speedups for quantum simulation
 - Simulating quantum physics and chemistry (Feynman's dream)
 - Probably the most important application of quantum computers so far!

$$N = pq$$

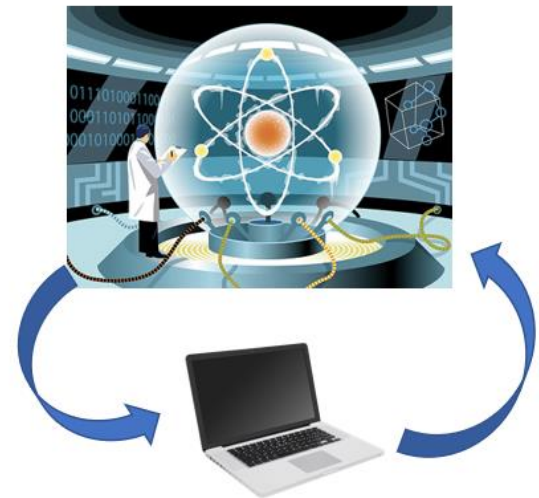


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Quantum algorithms

- Algorithms to be run on near-term QCs:
 - Variational eigensolvers
 - Classical-quantum hybrid algorithms
- Quantum Machine Learning
 - Linear system solvers/SDPs/Convex Optimization
 - Quantum neural nets
 - Recommendation systems
 - ...more

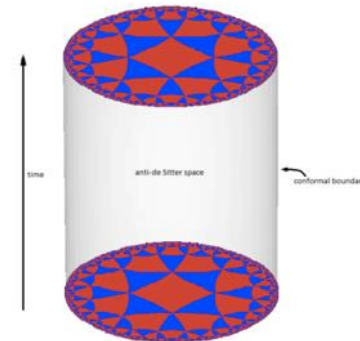


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Connections with fundamental physics

- The key to a theory of Quantum Gravity could be quantum entanglement, and quantum error correcting codes.
- The "Blackhole Firewall Paradox" is an issue about quantum information, and possible resolutions involve quantum cryptography.



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First quantum revolution:

Understanding quantum mechanics

1930



Birth of quantum information science:

Theoretical framework and foundational experiments for algorithms, communication, metrology

1980



Physics of Computation Conference Endicott House MIT May 6-8, 1981



2007:
First not-so-bad
qubit @ Yale

Second quantum revolution:

Applied use of quantum mechanics: information processing, acquisition, transmission and transduction

2020

2030

Fault-Tolerant
Quantum
Computing

ERA of the Noisy
Intermediate Scale
Quantum (NISQ) Systems

- First validations of Quantum Superiority
- First attempts to Quantum Advantage
- Foundations for scalable real world deployment of quantum-assisted technologies

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